

Exercise #8

6. March 2023

Problem 1. (Finite difference method for the wave equation)
 Consider the wave equation

$$\begin{aligned}
 u_{tt} &= 9u_{xx}, & -2 \leq x \leq 2, \quad t \geq 0, & \quad (1a) \\
 u(-2, t) &= u(2, t) = 0, & t \geq 0, & \quad (1b) \\
 u(x, 0) &= f(x), \quad u_t(x, 0) = g(x), & -2 \leq x \leq 2. & \quad (1c)
 \end{aligned}$$

a) Find the exact solutions of the wave equation for the following two sets of initial conditions:

- The conditions

$$f(x) = \sin\left(\frac{\pi x}{2}\right) \quad \text{and} \quad g(x) = 0. \quad (2)$$

- The conditions

$$f(x) = 0 \quad \text{and} \quad g(x) = \pi \sin\left(\frac{\pi x}{2}\right). \quad (3)$$

b) • Set up a finite difference scheme, using central differences for the second derivatives in both directions. Use step sizes h in space and k in the temporal direction. Notice that special care has to be taken for the first temporal step. The final scheme should be written in the form

$$\begin{aligned}
 U_i^1 &= \dots, & i &= 1, \dots, M-1, \\
 U_i^{n+1} &= \dots, & i &= 1, \dots, M-1, & n &= 1, 2, \dots,
 \end{aligned}$$

where the $\dots$ consist of given and/or previously computed values.

NB! Your solution should include an explanation on how the explicit scheme is obtained.

- Use the initial values (2) Let $h = 1$ and $k = \frac{1}{3}$ and use the formulas above to find approximations to $u(x_i, \frac{1}{3})$, $i = 1, \dots, 3$ (by hand!).
- c) Implement and test the scheme derived in point (b). A template for the implementation can be found in the Jupyternote `PDE_wave_fdm.ipynb`.

It is more interesting to study the animations of the simulations, however, for the hand-in, you are asked to plot the solutions for $t = 0, 0.4, 1.2$ and 2 if nothing else is said.

- Use the initial values from (2). Let $h = 0.1$ and $k = 0.01$. Plot the solutions, and compare with the exact solution.

You may also like to try (but not hand in) with k just a tiny bit larger than 0.01 , and observe what happens in this case. You should observe instabilities.

The next bullet points are optional and should not be handed in

- Repeat the experiment above with the initial values given by (3).
- Use the initial values (2), but let now the wave speed c be given by

$$c(x) = \begin{cases} 1 & \text{for } x \leq 0, \\ 2 & \text{for } x > 0. \end{cases}$$

Plot the solution for $t = 0, 0.4, 1.2$ and 2 .

Solution.

- a) For the solution of the wave equation on the bounded domain we can use the theory on the separation of variables developed at p. 545-547 since we are dealing with the same type of boundary conditions, so we do not repeat it here. *You are however expected to*

be able to perform the whole process outlined there without using the book. We have the general solution

$$F(x) = A \cos(px) + B \sin(px)$$

with k the separation constant and $k = -p^2$. In both conditions we have to guarantee:

$$F(-2) = F(2) = 0 \rightarrow \begin{cases} A \cos(p2) + B \sin(p2) = 0 \\ A \cos(-p2) + B \sin(-p2) = 0 \end{cases}$$

Using the symmetries of $\sin$, $\cos$ we can further develop

$$\begin{cases} A \cos(p2) + B \sin(p2) = 0 \\ A \cos(p2) - B \sin(p2) = 0 \end{cases}$$

Summing and subtracting these 2 equations we arrive at the equivalent requirement

$$2A \cos(p2) = 0 \quad 2B \sin(p2) = 0.$$

Since there is no point where $\cos$ and $\sin$ are zero at the same time we have the two cases

$$\begin{cases} B = 0 \\ p = (2n+1)\frac{\pi}{4}, \quad n = 0, \dots \end{cases} \quad \text{or} \quad \begin{cases} A = 0 \\ p = \frac{2n\pi}{4}, \quad n = 1, \dots \end{cases}$$

Having a look at the initial conditions imposed (but it can be verified testing the two cases) we choose the second option and introduce the index for the choice of the parameter n :

$$F_n(x) = B_n \sin(p_n x), \quad p_n = \frac{n\pi}{2}.$$

Adding the general solution for $k < 0$ of $G(t)$, we can arrive at (pg. 547):

$$u(x, t) = \sum_{n=0}^{\infty} [B_n \cos(\lambda_n t) + B_n^* \sin(\lambda_n t)] \sin(p_n x), \quad \lambda_n = 3p_n.$$

- Let us first study the case of initial values (2):

$$u(x, 0) = \sum_{n=0}^{\infty} B_n \sin(p_n x) = \sin \frac{\pi x}{2}$$

We can see that this is verified if $B_n = 0 \forall n \neq 1$ and $B_1 = 1$. In the same way

$$u_t(x, 0) = \sum_{n=0}^{\infty} \lambda_n B_n^* \sin(p_n x) = 0$$

which is true $\iff B_n^* = 0 \forall n$. The solution obtained is thus

$$u(x, t) = \cos(\lambda_1 t) \sin(p_1 x) = \cos\left(\frac{3\pi}{2}t\right) \sin\left(\frac{\pi}{2}x\right)$$

- The procedure is entirely analogous for the second case (2):

$$u(x, 0) = \sum_{n=0}^{\infty} B_n \sin(p_n x) = 0$$

This is verified if $\iff B_n = 0 \forall n$.

$$u_t(x, 0) = \sum_{n=0}^{\infty} \lambda_n B_n^* \sin(p_n x) = \pi \sin\left(\frac{\pi x}{2}\right)$$

which is true if $B_n^* = 0 \forall n \neq 1$ and $B_1^* = \frac{2}{3}$. The solution obtained is thus

$$u(x, t) = \frac{2}{3} \sin(\lambda_1 t) \sin(p_1 x) = \frac{2}{3} \sin\left(\frac{3\pi}{2}t\right) \sin\left(\frac{\pi}{2}x\right)$$

- b) We use central differences in space and time. We center our scheme at U_i^n and since the scheme is explicit all the quantities of the spatial derivative are referred to timestep t_n if we are computing the value at timestep t_{n+1} . We have:

$$\frac{U_i^{n+1} - 2U_i^n + U_i^{n-1}}{k^2} = 9 \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{h^2}. \quad (4)$$

rearranging and defining $r = 9k^2/h^2$ we have

$$U_i^{n+1} = rU_{i+1}^n + (2 - 2r)U_i^n + rU_{i-1}^n - U_i^{n-1}. \quad (5)$$

Since we have homogeneous Dirichlet boundary conditions the equations for the first and last point are:

$$U_1^{n+1} = rU_2^n + (2 - 2r)U_1^n + 0 - U_1^{n-1}, \quad (6)$$

$$U_{M-1}^{n+1} = 0 + (2 - 2r)U_{M-1}^n + rU_{M-2}^n - U_{M-1}^{n-1}. \quad (7)$$

Regarding the initial step, the values U_i^{-1} for $i = 1, \dots, M - 1$ are unknown. We can use the initial condition on the derivative and discretize it using first order central difference and thus imposing

$$\frac{U_i^1 - U_i^{-1}}{2k} = g(x_i) = 0 \rightarrow U_i^1 = U_i^{-1}. \quad (8)$$

The scheme for the initial timestep appears then in the form:

$$U_i^1 = rU_{i+1}^0 + (2 - 2r)U_i^0 + rU_{i-1}^0 - U_i^{-1} = rU_{i+1}^0 + (2 - 2r)U_i^0 + rU_{i-1}^0 - U_i^1 \quad (9)$$

$$\downarrow \quad (10)$$

$$2U_i^1 = rU_{i+1}^0 + (2 - 2r)U_i^0 + rU_{i-1}^0. \quad (11)$$

Regarding the first approximation using $h = 1$, $k = \frac{1}{3}$, we have $r = 1$ and

$$U_1^1 = 0.5 \cdot (1U_2^0 + (2 - 2 \cdot 1)U_1^0 + 1U_0^0) \quad (12)$$

$$= 0.5 \cdot (U_2^0 + U_0^0) \quad (13)$$

$$= 0.5 \cdot \sin(0) = 0 \quad (14)$$

$$U_2^1 = 0.5 \cdot (U_3^0 + U_1^0) = 0.5 \left(\sin\left(\frac{\pi}{2}\right) + \sin\left(\frac{-\pi}{2}\right) \right) = 0 \quad (15)$$

$$U_3^1 = 0.5 \cdot (U_4^0 + U_2^0) = 0.5 (\sin(\pi) + \sin(0)) = 0. \quad (16)$$

c) See JupyterNotebook PDE_wave_fdm_v2.ipynb

Problem 2.

Use D'Alembert's method to find the solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$$

for $t > 0$ and all $x \in \mathbb{R}$ with initial conditions

$$u(x, 0) = \sin x \quad \text{and} \quad \frac{\partial u(x, 0)}{\partial t} = e^{-x/4}.$$

Solution.

By definition the D'Alembert's solution is defined as

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds \quad (17)$$

where in our specific case we have:

$$f(x) = \sin(x), \quad g(x) = e^{-x/4}, \quad c = 2. \quad (18)$$

The solution is thus:

$$u(x, t) = \frac{1}{2} [\sin(x + ct) + \sin(x - ct)] + \frac{1}{4} \int_{x-ct}^{x+ct} e^{-s/4} ds \quad (19)$$

$$= \frac{1}{2} [\sin(x + 2t) + \sin(x - 2t)] + \frac{1}{4} \left[-4e^{-s/4} \right]_{x-2t}^{x+2t} \quad (20)$$

$$= \frac{1}{2} [\sin(x + 2t) + \sin(x - 2t)] - \left[e^{-(x+2t)/4} - e^{-(x-2t)/4} \right] \quad (21)$$

$$(22)$$

Problem 3. (1D Heat Equation)

Solve the differential equation

$$\begin{cases} \frac{\partial}{\partial t} u - \frac{1}{4} \frac{\partial^2}{\partial x^2} u = 0, \\ u(0, t) = u(\frac{\pi}{2}, t) = 0, \\ u(x, 0) = 2x(x - \frac{\pi}{2}) \end{cases}$$

for $x \in [0, \frac{\pi}{2}]$ and $t \in [0, \infty]$.

Solution.

By pages 560-561 in Kreyszig, this equation is solved by

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-n^2 t} \sin(2nx),$$

where B_n are the Fourier coefficients for the odd extension on $f(x) = 2x(x - \frac{\pi}{2})$. We compute

these coefficients:

$$\begin{aligned}
 B_n &= \frac{4}{\pi} \int_0^{\frac{\pi}{2}} 2x(x - \frac{\pi}{2}) \sin(2nx) dx \\
 &= \frac{4}{\pi} \left[2x(x - \frac{\pi}{2}) \frac{-\cos(2nx)}{2n} \right]_0^{\frac{\pi}{2}} - \frac{4}{\pi} \int_0^{\frac{\pi}{2}} 2(2x - \frac{\pi}{2}) \frac{-\cos(2nx)}{2n} dx \\
 &= \frac{4}{\pi} (0 - 0) + \frac{4}{n\pi} \int_0^{\frac{\pi}{2}} (2x - \frac{\pi}{2}) \cos(2nx) dx \\
 &= \frac{4}{n\pi} \left(\left[\frac{(2x - \frac{\pi}{2}) \sin(2nx)}{2n} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2 \frac{\sin(2nx)}{2n} dx \right) \\
 &= \frac{4}{n^2\pi} \left(0 - 0 - \int_0^{\frac{\pi}{2}} \sin(2nx) dx \right) \\
 &= -\frac{4}{n^2\pi} \left[-\frac{\cos(2nx)}{2n} \right]_0^{\frac{\pi}{2}} \\
 &= \frac{2}{n^3\pi} ((-1)^n - 1) = -\frac{4}{(2n+1)^3\pi}
 \end{aligned}$$

Finally:

$$u(x, t) = -\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^3} e^{-n^2 t} \sin(2nx)$$

The next exercises are optional and should not be handed in!

Problem 4.

Use D'Alembert's method to find the solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

for $t > 0$ and all $x \in \mathbb{R}$ with initial conditions

$$u(x, 0) = \cos x \quad \text{and} \quad \frac{\partial u(x, 0)}{\partial t} = xe^{-x^2}.$$

Solution.

By definition the D'Alembert's solution is defined as

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds \quad (23)$$

where in our specific case we have:

$$f(x) = \cos(x), \quad g(x) = xe^{-x^2}, \quad c = 1. \quad (24)$$

The solution is thus:

$$u(x, t) = \frac{1}{2} [\cos(x + t) + \cos(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} se^{-s^2} ds \quad (25)$$

$$= \frac{1}{2} [\cos(x + t) + \cos(x - t)] + \frac{1}{2} \left[\frac{e^{-s^2}}{-2} \right]_{x-t}^{x+t} \quad (26)$$

$$= \frac{1}{2} [\cos(x + t) + \cos(x - t)] - \frac{1}{4} [e^{-(x+t)^2} - e^{-(x-t)^2}] \quad (27)$$

$$(28)$$

Problem 5.

Vibrations in a stiff metal rod can be roughly described by the fourth order PDE

$$\frac{\partial^2 u}{\partial t^2} = -c^2 \frac{\partial^4 u}{\partial x^4}.$$

We assume now that the metal rod is simply supported at both ends at $x = 0$ and $x = \pi$, and has zero curvature at both ends. This corresponds to the boundary conditions

$$u(0, t) = u(\pi, t) = 0 \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2}(0, t) = \frac{\partial^2 u}{\partial x^2}(\pi, t) = 0$$

for all $t > 0$.

a) We use the idea of separation of variables and consider solutions of the form $u(x, t) = F(x)G(t)$. Derive ordinary differential equations for the functions F and G .

b) Verify that all functions of the form

$$F(x) = A \sin(\beta x) + B \cos(\beta x) + C \sinh(\beta x) + D \cosh(\beta x)$$

with $\beta > 0$ satisfy the ODE for F derived in the previous point. For which values of $\beta > 0$ are the boundary conditions satisfied? What are the corresponding solutions of the equation for G ?

Solution.

a) By definition we impose:

$$u(x, t) = F(x)G(t).$$

The PDE can thus be rewritten as:

$$F(x)\ddot{G}(t) = -c^2 F^{(4)}(x)G(t) \rightarrow \frac{F^{(4)}}{F} = -\frac{1}{c^2} \frac{\ddot{G}}{G} = k$$

where k is the separation constant. The system of ode's to solve is thus:

$$\begin{cases} F^{(4)}(x) - kF(x) = 0 \\ \ddot{G}(t) + kc^2 G(t) = 0 \end{cases} \quad (29)$$

b) If we differentiate $F(x)$ four times we obtain:

$$F'(x) = \beta[A \cos(\beta x) - B \sin(\beta x) + C \cosh(\beta x) + D \sinh(\beta x)] \quad (30)$$

$$F''(x) = \beta^2[-A \sin(\beta x) - B \cos(\beta x) + C \sinh(\beta x) + D \cosh(\beta x)] \quad (31)$$

$$F'''(x) = \beta^3[-A \cos(\beta x) + B \sin(\beta x) + C \cosh(\beta x) + D \sinh(\beta x)] \quad (32)$$

$$F^{(4)}(x) = \beta^4[A \cos(\beta x) - B \sin(\beta x) + C \cosh(\beta x) + D \sinh(\beta x)] = \beta^4 F(x) \quad (33)$$

Our function F thus solve the first ODE with $k = \beta^4$. If we want to apply the boundary conditions we have that these relations have to be satisfied by definition

$$u(0, t) = F(0)G(t) = 0 \quad (34)$$

$$u(\pi, t) = F(\pi)G(t) = 0 \quad (35)$$

$$u_{xx}(0, t) = F''(0)G(t) = 0 \quad (36)$$

$$u_{xx}(\pi, t) = F''(\pi)G(t) = 0 \quad (37)$$

Since we can't have $G(t) = 0$ otherwise we will have the trivial solution, F has to satisfy

$$F(0) = 0 \quad (38)$$

$$F(\pi) = 0 \quad (39)$$

$$F''(0) = 0 \quad (40)$$

$$F''(\pi) = 0 \quad (41)$$

which translates in

$$B + D = 0 \quad (42)$$

$$A \sin(\beta\pi) + B \cos(\beta\pi) + C \sinh(\beta\pi) + D \cosh(\beta\pi) = 0 \quad (43)$$

$$-B + D = 0 \quad (44)$$

$$\beta^2 [-A \sin(\beta\pi) - B \cos(\beta\pi) + C \sinh(\beta\pi) + D \cosh(\beta\pi)] = 0 \quad (45)$$

Taking the first and third statement it turns out that $B = D = 0$. What remains is

$$A \sin(\beta\pi) + C \sinh(\beta\pi) = 0 \quad (46)$$

$$\beta^2 [-A \sin(\beta\pi) + C \sinh(\beta\pi)] = 0 \quad (47)$$

equivalent to, since $\beta > 0$:

$$A \sin(\beta\pi) + C \sinh(\beta\pi) = 0 \quad (48)$$

$$-A \sin(\beta\pi) + C \sinh(\beta\pi) = 0. \quad (49)$$

Subtracting the second equation from the first we obtain $2A \sin(\beta\pi) = 0$. Since $\sinh$ has only one zero in $x = 0$ but choosing it would mean select the trivial solution, the only remaining possibility is choosing $C = 0$ and:

$$F_n(x) = A \sin(\beta_n x), \quad \beta_n = n = 1, \dots$$

The corresponding equation for G is $\ddot{G}(t) + n^4 c^2 G(t) = 0$ which has general solution:

$$G_n(t) = B_n \cos(\beta_n^2 ct) + B_n^* \sin(\beta_n^2 ct) \quad (50)$$

The general solution to the initial equation is thus

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \cos(\beta_n^2 ct) + B_n^* \sin(\beta_n^2 ct)) \sin(nx). \quad (51)$$